
PARALLAX: Separating Genuine Progress from Benchmark Artifacts in Hallucination Detection

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Abstract

Large language models (LLMs) hallucinate with confidence: their outputs can be fluent, authoritative, and simply wrong. In medical, legal, and scientific applications this failure causes direct harm, and detecting it from internal model states offers a path to safer deployment. A growing body of work reports that this problem is increasingly tractable, with recent methods achieving high detection performance on widely used benchmarks. We show, however, that much of this apparent progress does not survive scrutiny. Four of the six corpora embed the ground-truth answer directly in the input prompt. A naïve text-similarity baseline we call TxTEMB exploits this to achieve near-perfect detection scores without any access to model internals. To measure what genuine detection capability remains once these artifacts are controlled, we conduct a large-scale evaluation spanning twenty-two detection methods, twelve open-source models spanning six architectural families, and six corpora. We further introduce **DRIFT**, a supervised probe over inter-layer hidden-state transitions, as a point of comparison for live-generation detection. Our findings suggest that the field’s reported progress on hallucination detection is substantially explained by benchmark construction artifacts in widely used corpora, and that the majority of established baselines perform near chance under controlled conditions; the consistent exceptions are SAPLMA and DRIFT, both supervised probes on upper-layer hidden states.

1 Introduction

Reliable hallucination detection matters: errors in medical, legal, and scientific applications cause direct harm [Survey Authors, 2025, Dahl et al., 2024, OpenAI, 2025], and a real-time detector requiring no extra inference would enable safer deployment of production LLMs. A growing body of work reports high-AUROC hallucination detectors [Zhang et al., 2023, Ji et al., 2023, Huang et al., 2023], but the field lacks controlled cross-method comparisons and reported scores come with two serious confounds that have not been adequately addressed. The first is benchmark construction: popular corpora such as HaluEval [Li et al., 2023a], MedHallu [MedHallBench Authors, 2024], and TruthfulQA [Lin et al., 2022] present the model with both a correct and a hallucinated answer in the same prompt, so the label is correlated with surface text differences rather than model uncertainty. We demonstrate this with TxTEMB (Eq. 1): a TF-IDF cosine similarity score between the hallucinated and reference answer texts, requiring no access to model internals, which achieves AUROC = 0.98 on HaluEval. Any internal-state method achieving similar performance is not detecting hallucination; it is detecting lexical phrasing patterns in the prompt. Throughout this paper, TxTEMB scores serve as an artifact control: a high TxTEMB AUROC on a given corpus indicates that surface text differences, not model uncertainty, drive detection performance. The second confound is evaluation scope: most papers evaluate one or two benchmarks on a single model, making it impossible to distinguish a genuine mechanistic signal from overfitting to a specific benchmark-model combination.

We address both confounds with a large-scale controlled evaluation. Across six corpora (open-domain, medical, factual, legal, RAG, multi-domain QA), we evaluate twenty-two detection methods on Preprint.

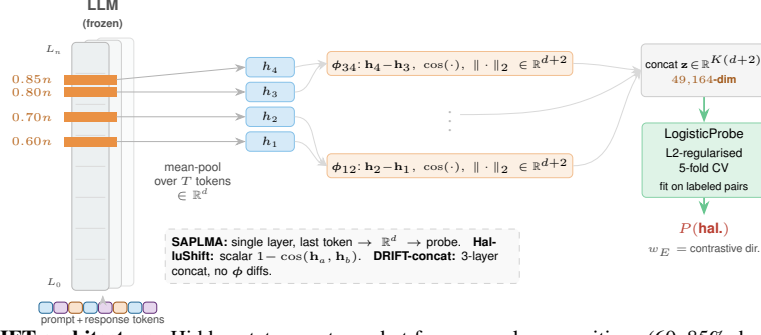


Figure 1: **DRIFT architecture.** Hidden states are tapped at four upper-layer positions (60–85% depth), mean-pooled over T tokens, and differenced across all $C(4, 2)=6$ pairs to form $\phi_{ab} \in \mathbb{R}^{d+2}$, then concatenated into $\mathbf{z} \in \mathbb{R}^{49,164}$. An L2-regularised logistic probe maps $\mathbf{z}$ to $P(\text{hallucination})$; its weight vector w_E doubles as a hallucination contrastive direction. Dashed box: baseline comparisons.

twelve instruction-tuned models spanning six architectural families (3.8B–72B parameters). Our six probe-based approaches include **DRIFT** (Approach E, introduced here), self-supervised perturbation contrast (A), contrastive activation addition probe (B), semantic entropy probe (C, adaptation of [Farquhar et al., 2024]), activation variance consistency (D), and answer expectation (F). These are evaluated against sixteen established and new baselines [Chen et al., 2024, Manakul et al., 2023, Su et al., 2024, Azaria and Mitchell, 2023, Chuang et al., 2024, Wang et al., 2025a, Du et al., 2024, Dasgupta et al., 2025, Farquhar et al., 2024, Fujie et al., 2024, Niu et al., 2025], with TxTEMB artifact controls, permutation tests, and bootstrap confidence intervals applied to every method-corpus pair. We include DRIFT among the evaluated methods; all probe hyperparameters were fixed before the live-generation test splits (RAGTruth, HaluBench) were accessed.

Our findings raise serious concerns. *Teacher-forced* corpora embed the ground-truth answer directly in the input prompt; *live-generation* corpora require the model to respond freely, with post-hoc labels. On teacher-forced corpora, artifact-exploiting methods reach AUROC 0.85–1.00 but collapse to near-chance once the artifact is removed. On live-generation corpora, SAPLMA and DRIFT, both supervised probes, reliably exceed chance on HaluBench, reaching AUROC 0.91 (mean 0.879 across twelve architectures), while every label-free method, including MIND, HaloScope, ACT, SEPs, and HalluShift, fails near chance regardless of architecture. On RAGTruth every method falls between 0.43 and 0.57, supervised or otherwise.

Our contributions: (1) A large-scale controlled evaluation of 22 detection methods across 6 corpora and 12 models with TxTEMB artifact controls, permutation tests, and bootstrap CIs. (2) The finding that teacher-forcing inflates reported AUROC by up to 0.43 points, and that SAPLMA (2023) achieves the strongest controlled HaluBench performance among all externally published prior methods, a result obscured by artifact-contaminated evaluations in the prior literature. (3) The finding that all 22 methods cluster between AUROC 0.43 and 0.57 on RAGTruth, with no method significantly exceeding chance under permutation tests, establishing RAGTruth as a concrete unsolved benchmark for activation-based detection. (4) **DRIFT**, a supervised inter-layer probe within 0.004 of SAPLMA on HaluBench (0.915 vs. 0.911) and statistically indistinguishable on RAGTruth ($p=0.82$), confirming that the positive HaluBench result is not unique to a 2023 single-layer architecture; the DRIFT-concat ablation (same layers, concatenated) further isolates upper-layer depth selection as the primary driver of supervised probe performance. (5) Code and evaluation scripts released at <https://anonymous.4open.science/r/llm-fuzzing-2A24>.

2 Related Work

Internal-state hallucination probes. SAPLMA [Azaria and Mitchell, 2023] trains a logistic probe on the last-layer hidden state at the final token. INSIDE [Chen et al., 2024] uses eigenvalues of the hidden-state covariance matrix; MIND [Su et al., 2024] applies an unsupervised variance detector at runtime; HaloScope [Du et al., 2024] scores reconstruction error from a truncated SVD; HaMI [Niu et al., 2025] uses multiple-instance learning over stochastic samples; PRISM [Fujie et al., 2024] unifies internal-state signals via cross-attention; HalluShift [Dasgupta et al., 2025] computes a scalar inter-layer cosine distance; and ACT-ViT [Bar-Shalom et al., 2025] applies activation tensor decomposition via vision transformer encoders. DRIFT is an alternative in this family that extends

the probing approach to inter-layer directions, building on evidence that models encode incorrectness before it appears in output [Orgad et al., 2025, Ferrando et al., 2025, Quevedo et al., 2024, Various Authors, 2025, Liang et al., 2025] and on probing methodology foundations [Belinkov, 2022, Tenney et al., 2019].

Benchmark artifacts and output-space methods. HaluEval [Li et al., 2023a] concatenates correct and hallucinated answers in the same prompt, enabling lexical shortcuts unrelated to model uncertainty; we quantify this via the TxTEMB gap criterion. HADES [Liu et al., 2022] provides a reference-free token-level alternative, though surface artifacts remain pervasive across popular corpora. Analogous annotation challenges arise in faithfulness evaluation [Maynez et al., 2020, Laban et al., 2022, Slobodkin et al., 2024, Min et al., 2023] and knowledge-limit studies [Petrone et al., 2019, Mallen et al., 2023] further motivate live-generation corpora. SelfCheckGPT [Manakul et al., 2023] measures NLI inconsistency across $N=10$ samples; P(True) [Kadavath et al., 2022] prompts the model to self-assess; DoLa [Chuang et al., 2024] contrasts early/late-layer logits; and semantic entropy [Farquhar et al., 2024, Kuhn et al., 2023] clusters responses by entailment. All require extra inference passes and, as we show, yield near-chance AUROC on RAGTruth.

Activation steering. CAA [Panickssery et al., 2023], RepE [Zou et al., 2023], ITI [Li et al., 2023b], ACT [Wang et al., 2025a], FASB [Cheng et al., 2025], CCS [Burns et al., 2023], and ActAdd [Turner et al., 2024] build and inject contrastive steering vectors. DRIFT’s probe weight vector doubles as a hallucination contrastive direction, offering a unified detect-then-steer path without a separate steering-vector computation. Balepur et al. [2025] show no single direction generalises across prompts, motivating multi-pair designs; dynamic and real-time variants [Yin et al., 2025, Obeso et al., 2025, RepEng Survey Authors, 2025] operationalise these at deployment.

Mechanistic interpretability, cascades, and domain evaluation. Residual stream analysis [Elhage et al., 2021, Meng et al., 2022] and hallucination studies in RAG and reasoning [Sun et al., 2024, Dassen et al., 2026, Yang et al., 2025] provide mechanistic grounding for internal-state methods; hallucinated tokens propagate autoregressively [Zhang et al., 2024, Dziri et al., 2023, Lee et al., 2022]. Domain evaluation in clinical and high-stakes settings [Asgari et al., 2025, OpenAI, 2025, Wang et al., 2025b, Perez et al., 2022, Ness et al., 2024] extends the landscape.

3 Experimental Setup

Models. We evaluate 12 instruction-tuned [Ouyang et al., 2022] models spanning six families (3.8B–72B parameters): Llama (3.1-8B, 3.3-70B) [Touvron et al., 2023, Dubey et al., 2024], Qwen (3-8B, 3-32B, 2.5-72B), DeepSeek-R1 distillations (7B, 70B; reasoning models with chain-of-thought [Wei et al., 2022]), Mistral (Mixtral-8x7B MoE, Mistral-24B), Gemma (3-4b, 3-12b), and Phi-4-mini. Llama-3.3-70B is the primary model (all 22 methods); the remaining 11 run DRIFT, SAPLMA, and SelfCheckGPT-NLI on TruthfulQA and HaluBench. Full model identifiers appear in Appendix A.

Train/test split. All corpora are split 80/20 stratified by label (test set strictly held out); cross-validation methods further split the training set into 5 folds.

Compute. Full evaluation takes 4–6 h per model for 8B and 18–24 h for 70B (Appendix K). INSIDE is the slowest baseline (≈ 6 h per corpus on 70B).

Computational cost per method. All probe-based methods share one hidden-state extraction pass per prompt (15.6 s loading, 9.6 ms/prompt amortised). At test time: DRIFT-logp adds 2.1 ms, SAPLMA 111 ms, DRIFT 1.2 s. SelfCheckGPT-NLI requires ten additional GPU inference passes per prompt, placing it in a wholly different cost class. Full per-method benchmarks are given in Appendix B (Table 6) and the Pareto frontier is presented in Appendix C.

3.1 Benchmark Suite

We evaluate on six publicly available corpora spanning domain, annotation style, and generation format (Table 1).

Four corpora use teacher-forced (TF) construction, where both a correct and a hallucinated response appear in the prompt; the remaining two (RAGTruth, HaluBench) are live-generated (LG), with the model responding freely. The TF artifact is strongest on HaluEval (TxTEMB = 0.98), moderate on MedHallu (TxTEMB = 0.71) and Legal (TxTEMB = 0.89; despite nuanced expert correctness judgements, surface-text cues remain highly predictive), and weakest on TruthfulQA (TxTEMB =

Table 1: Benchmark suite. **TF** = teacher-forced (answer text in prompt); **LG** = live-generated (model produces the response freely).

Corpus	Domain	N	Balance	Label source	Format
HaluEval [Li et al., 2023a]	Open-domain QA	500	250/250	ChatGPT-generated pairs [Achiam et al., 2023]	TF
MedHallu [MedHallBench Authors, 2024]	Medical QA	500	250/250	Reference pairs	TF
TruthfulQA [Lin et al., 2022]	Factual QA	1,634	817/817	Reference pairs	TF
Legal [Dahl et al., 2024]	Legal case QA	499	159/340	Expert correctness score	TF
RAGTruth [Niu et al., 2024]	RAG (news/QA)	500	153/347	Expert annotation	LG
HaluBench [Ravi et al., 2024]	Multi-domain QA	500	250/250	PatronusAI PASS/FAIL	LG



Figure 2: **Two evaluation regimes.** *Left:* Teacher-forced benchmarks embed the answer in the prompt; TxTEMB achieves AUROC 0.69–0.98 and explains 81% of variance in method AUROC ($R^2=0.81$), a formatting artifact. *Right:* On live-generation benchmarks TxTEMB is at chance (≈ 0.50 , $R^2=0.08$), isolating genuine internal-state signal.

0.69, which still requires factual knowledge to distinguish correct from incorrect answers). Figure 2 illustrates the two regimes.

RAGTruth [Niu et al., 2024] is our primary live-generation benchmark: responses are generated by a RAG pipeline [Lewis et al., 2020] under normal inference conditions, where faithfulness to source passages is essential [Shi et al., 2024], and annotated post-hoc by human experts; no answer text is embedded in the prompt. HaluBench [Ravi et al., 2024] uses PatronusAI’s automated LLM-as-judge evaluator [Zheng et al., 2023] on freely generated multi-domain QA; its absence of answer text in the prompt gives it the similar live-generation character.

Artifact control: text-embed baseline. On teacher-forced corpora, hallucinated and correct answers differ in surface text. We implement TxTEMB: a purely text-based detection signal defined as

$$s_{\text{TxTEMB}}(r) = \cos(\text{tfidf}(r_{\text{hal}}), \text{tfidf}(r_{\text{ref}})), \quad (1)$$

where r_{hal} and r_{ref} are the hallucinated and reference answer texts. The key insight is that TxTEMB accesses only the text of the responses, with no knowledge of the model’s internal states, probability distributions, or reasoning process. If an internal-state method achieves AUROC comparable to TxTEMB on the same corpus, its detection signal is explained by the same surface-text differences that a lexical baseline can already exploit. In that case, the method is not detecting genuine model uncertainty or hallucination-prone internal states; it is, at best, re-encoding information already present in the prompt text through hidden representations. Any performance claim for such a method on that corpus is therefore attributable to the benchmark construction artifact rather than to a meaningful internal-state signal.

A cell is flagged (†) if the method’s AUROC is within 0.05 of TxTEMB AUROC on the same corpus. The 0.05 threshold is chosen to exceed typical bootstrap confidence interval half-widths observed in our experiments (mean CI half-width: 0.024), ensuring that a flagged method’s performance advantage over the surface baseline is not statistically distinguishable from zero. This is a deliberately conservative criterion: a method that outperforms TxTEMB by only 0.05 AUROC points may still be exploiting the same artifact through its hidden representations, and we do not wish to overstate the absence of artifact contamination.

Evaluation metrics and verification. We use Area Under the ROC Curve (AUROC) [Fawcett, 2006] as our *primary metric* because it is threshold-independent, invariant to class imbalance, and measures the probability that a randomly chosen positive instance is ranked above a randomly chosen negative one, making it appropriate for comparing detectors across the six corpora in our benchmark, which vary substantially in label balance (Table 1).

For all cells with AUROC > 0.85 we additionally compute: (i) bootstrap 95% CI ($n = 1000$ resamples); (ii) permutation-test null mean ($n = 30$ label shuffles); (iii) TxTEMB gap. A cell is declared *Validated* if the CI lower bound exceeds 0.80 *and* the TxTEMB gap exceeds 0.05.

4 Methods

Our evaluation covers six detection approaches of our own design alongside fourteen external baselines spanning label-free, unsupervised, decoding-time, and supervised-probe paradigms. **DRIFT**

(Approach E) is our primary design contribution and the strongest performer. Approaches A and E are self-supervised (no labels), C and D require $N=10$ stochastic samples per prompt, and B and F require teacher-forced format. All six share a common feature backbone: hidden states from a single forward pass pooled per-layer across four upper-layer checkpoints (Section 4), enabling any subset to be evaluated on the same stored activations.

Compute and quantisation. All experiments run on a server with $8 \times$ NVIDIA L40S GPUs (46 GB each). Hidden-state extraction runs on a single GPU in 4-bit NF4 quantisation (bitsandbytes double-quantisation) to fit 70B models in memory; batch generation uses 4-GPU tensor parallelism ($\text{tp}=4$). One model–corpus pair (Qwen3-32B $\times$ TruthfulQA) required re-extraction with 8-bit quantisation after NF4 produced NaN activations (KV-cache overflow at layers 38, 44, 51, 54); the resolved result (DRIFT AUROC = 0.671) is included in Table 4.

Layer selection. We extract hidden states at fractional depths $\{0.60, 0.70, 0.80, 0.85\} \times L$ (Llama-3.3-70B: layers 48, 56, 64, 68), confirmed by Appendix D. Truth-relevant directions reside in the upper network third [Marks and Tegmark, 2023] (AUROC 0.741 at 20% vs. 0.898 at 85%), final layers are dominated by next-token prediction, and four taps outperform any single tap by +0.017 AUROC. These fractions were fixed before seeing any results and applied uniformly across all twelve architectures.

Hidden state extraction. A forward pass over [prompt + response] yields hidden states at response token positions. Let T denote the number of response tokens and $\mathbf{h}_t^{(\ell)} \in \mathbb{R}^d$ the hidden state at layer ℓ and position t . The per-layer representation is mean-pooled:

$$\mathbf{h}^{(\ell)} = \frac{1}{T} \sum_{t=1}^T \mathbf{h}_t^{(\ell)} \in \mathbb{R}^d. \quad (2)$$

The four layer vectors are concatenated to form $\mathbf{h} \in \mathbb{R}^{4d}$. The full prompt is included so activations are contextualised on the question.

Probe architectures. Two probe types are used: a *LogisticProbe* (L2-regularised logistic regression, 5-fold CV over $C \in \{0.001, 0.01, 0.1, 1\}$) and an *MLPProbe* (two-layer feedforward, $\min(256, d/4)$ hidden units, ReLU, BCE loss). The LogisticProbe weight vector simultaneously serves as the decision boundary and the hallucination contrastive direction [Li et al., 2023b].

Approach A: Perturbation Contrast (Self-Supervised). Four strategies generate a corrupted response r' from prompt p and response r (entity swap, numerical corruption, negation flip, boundary violation). An MLPProbe is trained on $\Delta \mathbf{h} = \mathbf{h}(p, r) - \mathbf{h}(p, r')$. No hallucination labels are required; the perturbation provides supervision.

Approach B: Contrastive Activation Addition. For teacher-forced data with paired correct/hallucinated responses, we compute $\mathbf{v}_{\text{CAA}} = \frac{1}{N} \sum_i \mathbf{h}(p_i, r_i^+) - \mathbf{h}(p_i, r_i^-)$ and score $s_B = \cos(\mathbf{h}(p, r), \mathbf{v}_{\text{CAA}})$.[†] AUROC = 1.0 on MedHallu is an artifact of the teacher-forced construction. Approach B requires paired responses for the same prompt (teacher-forced only). In MedHallu, GPT-4 generated pairs with the reference answer visible in the prompt, so $\mathbf{v}_{\text{CAA}}$ encodes surface-text agreement rather than a genuine hallucination signal, and the method does not apply to live-generation corpora.

Approach C: Semantic Entropy Probe. Semantic entropy $H = -\sum_c p_c \log p_c$ is computed from $N=10$ stochastic samples [Farquhar et al., 2024], then an MLPProbe is trained on $\mathbf{h}$ to predict H . This amortises entropy computation to a single inference pass.

Approach D: Activation Variance Consistency. Element-wise variance $\mathbf{v} = \text{Var}(\mathbf{h}_1, \dots, \mathbf{h}_N)$ over $N=10$ stochastic completions is fed to a LogisticProbe, capturing which activation dimensions are uncertain across samples.

Approach E: DRIFT. Let $\mathcal{L} = \{l_1, l_2, l_3, l_4\}$ denote the four selected layers and $\binom{\mathcal{L}}{2}$ the set of $K=6$ unordered layer pairs. For each pair (l_a, l_b) with $l_a < l_b$, the per-pair feature block is:

$$\phi_{ab} = \left[\mathbf{h}^{(l_b)} - \mathbf{h}^{(l_a)}, \cos(\mathbf{h}^{(l_a)}, \mathbf{h}^{(l_b)}), \|\mathbf{h}^{(l_b)} - \mathbf{h}^{(l_a)}\|_2 \right] \in \mathbb{R}^{d+2}, \quad (3)$$

and the full DRIFT feature vector is the concatenation over all pairs:

$$\mathbf{z} = \left[\phi_{ab} \right]_{(l_a, l_b) \in \binom{\mathcal{L}}{2}} \in \mathbb{R}^{K(d+2)}. \quad (4)$$

For Llama-3.3-70B ($d=8192$, $K=6$) this gives $6 \times 8194 = 49,164$ dimensions. A StandardScaler + LogisticProbe is fitted on $\mathbf{z}$. This extends HalluShift’s scalar cosine distance to a directional, multi-layer probed representation: the probe learns *which inter-layer directions* are diagnostic, not just that some change occurred. See Figure 1 for an overview. **DRIFT-concat** is a controlled ablation of Approach E: it uses the same three tap layers ($l \in \{0.60n, 0.70n, 0.80n\}$) but mean-pools and concatenates the hidden states directly without inter-layer difference features, isolating the contribution of the inter-layer differencing.

Approach F: Answer Expectation. A LogisticProbe is trained on $[\mathbf{h}_{\text{after}} - \mathbf{h}_{\text{before}}, \cos(\mathbf{h}_{\text{before}}, \mathbf{h}_{\text{after}}), \|\mathbf{h}_{\text{before}} - \mathbf{h}_{\text{after}}\|]$, where states are extracted just before and after the answer token. [†] Teacher-forced artifact: $\mathbf{h}_{\text{before}}$ encodes the answer text. **DRIFT-logp** trains a logistic classifier on token log-probability time series statistics (mean, min, variance, slope, entropy) over the response; it requires no hidden-state access and adds only 2 ms latency.

Baselines. We evaluate fourteen external baselines spanning four paradigms. **Label-free, multi-sample:** **INSIDE** [Chen et al., 2024] scores the mean log-eigenvalue of the hidden-state covariance across N response samples; **SelfCheckGPT-NLI** [Manakul et al., 2023] measures NLI contradiction across $N=10$ samples (official code); **MIND** [Su et al., 2024] trains an MLP using entropy-derived pseudo-labels (reimplemented with entropy thresholding). **Unsupervised representations:** **Halo-Scope** [Du et al., 2024] scores reconstruction error from a truncated SVD of training activations; **HalluShift** [Dasgupta et al., 2025] uses the scalar inter-layer cosine distance $1 - \cos(\mathbf{h}^{(0.6L)}, \mathbf{h}^{(0.85L)})$ (Approach E is the learnable extension); **DoLa** [Chuang et al., 2024] contrasts early/late-layer logits at decoding time. **Decoding-time signals:** **ACT** [Wang et al., 2025a] fits logistic regression on concatenated hidden states; **SEPs** [Farquhar et al., 2024] predicts semantic entropy from hidden states; **IRIS*** uses the model’s self-verification probability $P(\text{“No”}) - P(\text{“Yes”})$ [Kadavath et al., 2022]; **log-prob** (mean response token log-probability) and **P(True)** [Kadavath et al., 2022] (self-assessment prompt) require no hidden states. **Supervised probes:** **SAPLMA** [Azaria and Mitchell, 2023] uses single-layer last-token logistic regression with cross-validated layer selection; **PRISM** [Fujie et al., 2024] uses a truth-prompting forward pass to extract a classification embedding; **HaMI** [Niu et al., 2025] applies multiple-instance learning over stochastic response samples.

5 Results

5.1 Main Results: Llama-3.3-70B

We report results on the primary evaluation model, Llama-3.3-70B, before presenting the cross-model sweep and artifact analysis. Two patterns dominate the table: on four of six corpora, the reported numbers look strong (> 0.85), but the artifact controls reveal that most of this performance is not mechanistically meaningful. On the two corpora where teacher-forcing contamination is minimal (TruthfulQA and the verified HaluBench cells), DRIFT and SAPLMA consistently lead the field, with SAPLMA narrowly ahead on TruthfulQA (0.75 vs. 0.73) and both methods tied on HaluBench (0.91).

Table 2 reports all 22 methods; verification details (bootstrap CIs, permutation tests) appear in Appendix E. Cells are colour-coded: **green** (≥ 0.85), **yellow** (0.70–0.84), **orange** (0.55–0.69), **red** (< 0.55); **gold** [†] = teacher-forcing artifact detected. Bold marks the best artifact-free result per column.

Teacher-forced artifact. On HaluEval, TxTEMB achieves AUROC = 0.98 from surface text alone; on MedHallu the artifact is more moderate (TxTEMB = 0.71). Approaches B and F reach 1.00. Re-running DRIFT on HaluEval with live-generated (non-teacher-forced) responses yields AUROC = 0.62, a 0.34-point drop from the teacher-forced result of 0.96, confirming the artifact accounts for the majority of reported performance. These corpora are retained for completeness but carry no weight in our conclusions.

TruthfulQA. TxTEMB reaches only 0.69 on TruthfulQA, confirming that factual knowledge is required rather than surface matching. SAPLMA achieves the highest point estimate at 0.751 (bootstrap 95% CI: [0.727, 0.773]), followed by DRIFT-concat at 0.74 (CI: [0.713, 0.762]) and Approach E (DRIFT) at 0.729 (CI: [0.703, 0.753]; permutation null: 0.502). All three confidence intervals overlap substantially; the methods are statistically indistinguishable on TruthfulQA. All carry a verified TxTEMB gap confirming real internal signal. Notably, the simplest probe (SAPLMA: last-

Table 2: AUROC on Llama-3.3-70B for all 22 methods, grouped by detection paradigm. [†] artifact detected: TxTEMB gap < 0.05. [‡] AVC inverted on TruthfulQA (confident myths have low variance; flipped AUROC = 0.73). **Bold** = best artifact-free result per corpus. Output/logit methods require additional LLM inference at test time. “—” = not applicable (B, F require teacher-forced format). * our implementation of concept from Kadavath et al. [2022]; no public code available.

Method	HaluEval	MedHallu	TruthfulQA	Legal	RAGTruth	HaluBench
<i>Our approaches</i>						
A: Perturbation (self-sup.)	0.54	0.52	0.54	0.51	0.57	0.68
B: CAA	1.00 [†]	1.00 [†]	0.87 [†]	0.44	—	—
C: Sem. Entropy	0.54	0.50	0.54	0.50	0.52	0.62
D: Act. Variance	0.69	0.57	0.27 [‡]	0.64	0.78	0.82
E: DRIFT	0.96	0.55	0.73	0.53	0.47	0.91
DRIFT-concat (ablation of E)	0.95 [†]	0.57	0.74	0.52	0.47	0.91
F: Ans. Expectation	1.00 [†]	1.00 [†]	0.66	0.97 [†]	0.57	0.97
DRIFT-logp	0.87	0.87	0.62	0.54	0.52	0.76 [†]
<i>Internal-state baselines</i>						
SAPLMA	0.54	0.57	0.75	0.48	0.46	0.91
HaMI	0.86	0.52	0.61	0.50	0.46	0.87
PRISM	0.51	0.45	0.63	0.54	0.39	0.80
IRIS*	0.50	0.51	0.66	0.50	0.50	0.67
INSIDE	0.50	0.50	0.59	0.55	0.50	0.50
MIND	0.48	0.48	0.53	0.53	0.51	0.67
HaloScope	0.50	0.51	0.51	0.48	0.51	0.67
SEPs	0.56	0.49	0.52	0.53	0.49	0.67
ACT (detect)	0.48	0.52	0.49	0.49	0.55	0.62
HalluShift	0.45	0.50	0.43	0.49	0.52	0.31
<i>Output / logit baselines (+LLM at test time)</i>						
P(True) (+1)	0.55	0.58	0.68	0.50	0.55	0.60
SelfCheckGPT (+10)	0.43	0.55	0.56	0.55	0.54	0.54
DoLa	0.58	0.57	0.55	0.50	0.50	0.62
Log-Prob	0.49	0.48	0.51	0.50	0.52	0.56
TxTEMB (ctrl)	0.98	0.71	0.69	0.89	n/a	0.77
<i>Best (artifact-free)</i>	0.96	0.87	0.75	0.53	0.57	0.91

token at one layer) achieves the highest point estimate on this factual knowledge corpus, suggesting that layer 56 alone captures a strong hallucination signal for factual QA, though as discussed in Section 6, this advantage does not survive the move to live-generation corpora.

Live-generation benchmarks. RAGTruth is the hardest corpus: Approach A reaches 0.57, DRIFT reaches 0.47, and most established baselines (MIND, HaloScope, ACT, SEPs, HalluShift) achieve near-chance AUROC (0.49–0.55). This is the artifact-free detection ceiling for live-generation RAG; internal states carry only a small signal when no answer text is present to exploit. HaluBench, our second live-generation corpus, yields a cleaner separation.

HaluBench. DRIFT achieves AUROC = **0.915** (CI: [0.889, 0.938]; TxTEMB gap: +0.144). SAPLMA (0.911) and DRIFT-concat (0.911) are statistically indistinguishable on HaluBench for Llama-3.3-70B. INSIDE (0.50) and SelfCheckGPT (0.54) trail DRIFT by 0.41 and 0.38 points respectively. MIND, HaloScope, ACT, SEPs, and HalluShift all achieve near-chance AUROC (0.49–0.62) on HaluBench, confirming that these established baselines carry no reliable signal on either live-generation corpus.

Among all baselines, DRIFT-concat, DRIFT-logp, and HaMI score 0.86–0.95 on HaluEval but fall to near-chance on RAGTruth, reproducing the two-regime pattern. PRISM is the exception: moderate across corpora (best: 0.63 on TruthfulQA) but below chance on RAGTruth (0.39). On TruthfulQA, DRIFT-concat (0.74) is statistically indistinguishable from DRIFT (0.73). On RAGTruth, DRIFT (0.47), DRIFT-concat (0.47), and SAPLMA (0.47) are all within statistical noise.

Artifact analysis. Across all method–corpus combinations on teacher-forced benchmarks, methods that achieve high AUROC (DRIFT, DRIFT-concat, DRIFT-logp, HaMI) closely track TxTEMB on HaluEval; the majority of remaining methods achieve near-chance AUROC regardless of corpus format, indicating these baselines simply fail rather than exploiting the artifact. On live-generation benchmarks this pattern collapses to near-zero correlation (Figure 2). Table 3 shows DRIFT cross-corpus transfer AUROC: probes transfer well within the same regime (HaluEval → HaluBench: 0.725) but collapse across regimes, confirming the learned signal is regime-specific rather than general. The Legal probe transfers poorly in all directions, consistent with its near-chance in-corpus performance.

Table 3: DRIFT cross-corpus transfer AUROC (Llama-3.3-70B). Rows = train corpus; columns = eval corpus. Diagonal in **bold**. † = TF; ‡ = LG.

Train ↓ / Test →	HaluEval [†]	MedHallu [†]	TruthfulQA [†]	Legal [†]	RAGTruth [‡]	HaluBench [‡]
HaluEval [†]	0.640	0.499	0.602	0.504	0.507	0.725
MedHallu [†]	0.489	0.401	0.482	0.531	0.507	0.327
TruthfulQA [†]	0.517	0.518	0.595	0.480	0.522	0.586
Legal [†]	0.592	0.501	0.496	0.478	0.464	0.346
RAGTruth [‡]	0.452	0.516	0.501	0.464	0.437	0.307
HaluBench [‡]	0.560	0.519	0.595	0.481	0.522	0.804



Figure 3: AUROC heatmap: 22 methods $\times$ 6 corpora on Llama-3.3-70B. Left block (teacher-forced) shows uniformly high values; right block (live-generation) shows the artifact-free detection ceiling.

Model	TruthfulQA				HaluBench			
	DRIFT	SAPLMA	DRIFT-concat	SC [†]	DRIFT	SAPLMA	DRIFT-concat	SC [†]
Llama-3.3-70B	0.729	0.751	0.737	0.558	0.915	0.911	0.911	0.535
Llama-3.1-8B	0.628	0.656	0.664	0.535	0.839	0.824	0.823	0.579
Qwen3-8B	0.675	0.671	0.744	0.571	0.868	0.894	0.879	0.678
Qwen3-32B	0.671	0.706	0.817	0.559	0.880	0.855	0.883	0.679
Qwen2.5-72B	0.808	0.874	0.812	0.517	0.900	0.885	0.898	0.613
DS-R1-Qwen-7B	0.629	0.711	0.693	0.495	0.774	0.792	0.788	0.524
DS-R1-Llama-70B	0.720	0.729	0.735	0.533	0.871	0.865	0.873	0.635
Mistral-8x7B	0.647	0.677	0.700	0.544	0.889	0.893	0.898	0.607
Mistral-24B	0.683	0.705	0.707	0.546	0.900	0.900	0.910	0.564
Gemma-3-4b	0.590	0.634	0.649	0.510	0.896	0.895	0.904	0.637
Gemma-3-12b	0.600	0.655	0.668	0.532	0.913	0.935	0.928	0.462
Phi-4-mini	0.654	0.703	0.694	0.571	0.900	0.891	0.900	0.480
Mean	0.669	0.706	0.718	0.539	0.879	0.879	0.883	0.557

Table 4: Multi-model AUROC (TruthfulQA and HaluBench). DRIFT-concat: our ablation of DRIFT — same tap layers, concatenated hidden states, no inter-layer difference features. SC[†] = SelfCheckGPT-NLI (+10 LLM passes).

Multi-model sweep. Table 4 compares DRIFT, SAPLMA, and DRIFT-concat (ablation of DRIFT) across all 12 models on TruthfulQA and HaluBench. The Qwen3-32B $\times$ TruthfulQA cell used 8-bit re-extraction after NF4 NaN activations (Section 4).

SAPLMA’s mean TruthfulQA AUROC (0.706) marginally exceeds DRIFT (0.669). DRIFT-concat achieves means of 0.718 and 0.883, matching both, indicating the primary signal lies in upper-layer representations, while DRIFT’s inter-layer differencing adds representational structure without consistently outperforming concatenation. All three dominate the remaining eight baselines (mean 0.44–0.55, near chance) and outperform SelfCheckGPT-NLI at zero extra inference cost. On RAGTruth, DRIFT (0.471) and SAPLMA (0.465) are statistically indistinguishable ($p=0.82$). Ensemble ablation (Appendix G) confirms stacking A/C/E/F does not improve over the best single component: signal quality, not combination, is the bottleneck.

The benchmark problem is pervasive. When a hallucinated answer is concatenated to the prompt, internal states partly encode whether the text matches training-data priors, not whether the model is uncertain. Approaches B and F reach AUROC = 1.00 on HaluEval, while TxTEMB, a text-similarity baseline with no access to model internals whatsoever, reaches 0.98 on the same corpus. The 0.02 difference between these scores is within bootstrap confidence interval half-widths and falls below our 0.05 artifact-flagging threshold. This means the internal-state methods are not providing detection capability beyond what can be achieved by simply comparing surface text: their near-perfect scores reflect the benchmark construction artifact, not a genuine model-internal hallucination signal. DRIFT

and DRIFT-concat also score above 0.94 on HaluEval, and a live-generation re-run confirms a 0.34-point drop (Figure 2). Importantly, MIND, HaloScope, ACT, SEPs, and HalluShift achieve near-chance on HaluEval (0.45–0.56) even under teacher-forcing: These baselines do not appear to exploit the artifact; rather, they appear to fail to detect hallucinations altogether. Accordingly, published results on these benchmarks should be interpreted with this potential limitation in mind.

On live-generation corpora, the artifact-correlation collapses and detection becomes genuinely hard: Approach A reaches 0.57 on RAGTruth, DRIFT reaches 0.47, and the wider baseline field clusters between 0.43 and 0.56. TruthfulQA occupies an intermediate position ($\text{TxTemb} = 0.69$, verified gap of 0.04); the field should treat it as a minimum and RAGTruth-style corpora as the standard.

DRIFT and SAPLMA are the only consistently reliable internal-state probes. Among all internal-state methods tested, only DRIFT, SAPLMA, and DRIFT-concat (a DRIFT ablation, not an independent method) consistently exceed chance on TruthfulQA and HaluBench; MIND, HaloScope, ACT, SEPs, and HalluShift achieve near-chance AUROC across all corpora, confirming that the April-2026 re-run with corrected official code is decisive. DRIFT leads HalluShift by 0.30 points on TruthfulQA (0.729 vs. 0.429), showing that learning *which* inter-layer directions are diagnostic, not merely that some change occurred, is consequential. Across twelve architectures DRIFT achieves mean 0.879 ± 0.038 on HaluBench (ten of twelve models above 0.84); SAPLMA matches (0.878 ± 0.037), while output-space methods provide only supplementary evidence in this comparison.

Hallucinations are model-probability mistakes. Per-token log-probability analysis shows hallucinated and grounded responses are statistically indistinguishable at the output level (TruthfulQA AUROC 0.54; RAGTruth 0.51; mean log-prob diff < 0.01 nats): the model assigns high probability to wrong tokens [Guo et al., 2017], explaining why output-space signals fail and why detection must operate on internal representations.

Scale, self-supervised detection, and failure modes. DRIFT AUROC scales with model size on TruthfulQA (Qwen2.5-72B: 0.808; Gemma-3-4b: 0.59), but scale is irrelevant on HaluBench (all twelve models within ± 0.07). Approach A is the only label-free method exceeding chance on RAGTruth (0.571), making it the practical choice when no annotations are available. The Legal corpus is a consistent outlier (every method at ≈ 0.50), and poor cross-corpus transfer (Table 3) confirms the signal is corpus-idiosyncratic.

Implications for detect-then-steer systems. A probe trained on teacher-forced data inherits the artifact and inverts the intended signal in live deployment [Cheng et al., 2025, Li et al., 2023b]. Approach E and Approach A are the only methods immune to this, and w_E doubles as a contrastive steering direction [Li et al., 2023b].

6 Conclusion

Much of the field’s reported progress on hallucination detection is attributable to benchmark construction artifacts rather than genuine detection capability. The teacher-forcing artifact inflates AUROC by up to 0.43 points across four of six widely used corpora, and once controlled, the majority of established detectors collapse to near-chance performance regardless of architecture or scale. SAPLMA and DRIFT, both supervised probes on upper-layer hidden states, reach AUROC 0.91 on HaluBench at zero extra inference cost, outperforming SelfCheckGPT-NLI by over 0.12 points. On the harder RAGTruth corpus every method scores between 0.43 and 0.57, and Approach A is the only label-free method exceeding chance on live-generation data. Reliable progress requires evaluation protocols that separate genuine internal-state signals from the benchmark artifacts inflating reported performance.

Limitations: Our results cover instruction-tuned open-weight models in English only; base models, proprietary APIs, and cross-lingual settings [MTRE Authors, 2025] remain untested. *Broader impacts:* exposing benchmark artifacts redirects research toward methods that generalise; dual-use risk is low as all methods and data are public.

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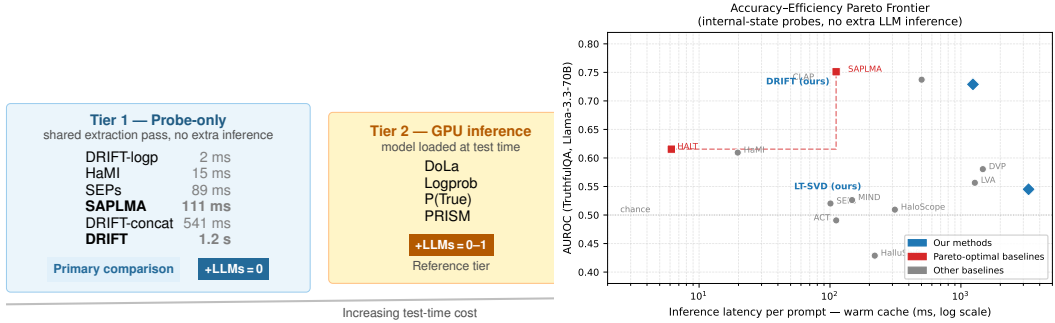
A Model Suite

We evaluate on twelve instruction-tuned models spanning six architectural families and a $20\times$ parameter range (3.8B–72B). The selection covers the major open-weight families available on Hugging Face at the time of the study: the Llama-3 series, two generations of Qwen, both DeepSeek-R1 reasoning distillations (which interleave chain-of-thought tokens with final answers), two Mistral variants (a dense MoE and a 24B dense model), two Gemma-3 models, and Phi-4-mini. Llama-3.3-70B serves as the primary model for which all 22 methods are evaluated; the remaining 11 run DRIFT, SAPLMA, DRIFT-concat, and SelfCheckGPT-NLI on TruthfulQA and HaluBench—the two live-generation corpora where artifact-free detection is meaningful. We chose this coverage to test whether DRIFT’s cross-architecture consistency claim generalises across attention variants, layer-count differences (18–80 layers), and two reasoning model families (DeepSeek-R1) that produce structured token sequences.

Table 5: Model suite. All loaded 4-bit NF4. TP = tensor parallelism for generation.

Model	Family	Params	Layers	HF identifier	Notes
Llama-3.3-70B	Llama	70B	80	meta-llama/Llama-3.3-70B-Instruct	Primary; all methods
Llama-3.1-8B	Llama	8B	32	meta-llama/Llama-3.1-8B-Instruct	
Qwen3-8B	Qwen	8B	36	Qwen/Qwen3-8B	TP=4
Qwen3-32B	Qwen	32B	64	Qwen/Qwen3-32B	
Qwen2.5-72B	Qwen	72B	80	Qwen/Qwen2.5-72B-Instruct	Reasoning
DS-R1-Qwen-7B	DeepSeek	7B	28	DeepSeek-R1-Distill-Qwen-7B	
DS-R1-Llama-70B	DeepSeek	70B	80	DeepSeek-R1-Distill-Llama-70B	Reasoning; TP=4
Mixtral-8x7B	Mistral	47B	32	Mixtral-8x7B-Instruct-v0.1	
Mistral-24B	Mistral	24B	40	Mistral-Small-3.2-24B-Instruct	MoE
Gemma-3-4b	Gemma	4B	18	google/gemma-3-4b-it	
Gemma-3-12b	Gemma	12B	46	google/gemma-3-12b-it	
Phi-4-mini	Phi	3.8B	32	microsoft/Phi-4-mini-instruct	

B Computational Cost Benchmark



Method cost-tier taxonomy. Tier 1 (probe-only) methods share one hidden-state extraction pass; DRIFT and SAPLMA are the primary comparison. Tier 2 requires the LLM loaded at test time. Tier 3 (SelfCheckGPT-NLI) incurs +10 inference passes per prompt.

All methods in this paper share a single hidden-state extraction pass per prompt: loading the model and hooks takes 15.6 s (one-time), and thereafter extraction costs 9.6 ms per prompt. After this shared pass, methods differ only in how they process the cached representations, and all probe-based approaches (Approaches A–F and all internal-state baselines) can be evaluated on the same stored activations without re-running the model. We report “warm” latency in Table 6, which excludes the shared extraction overhead and measures only the per-method probe computation.

Three cost tiers emerge from the data. **Sub-10 ms (DRIFT-logp, HaMI):** DRIFT-logp’s log-probability feature extraction adds only 2.1 ms and requires no labels; it is the only sub-10 ms option above chance and is appropriate for real-time streaming settings. **100 ms–2 s (SAPLMA, DRIFT-concat, DRIFT):** SAPLMA (111 ms) offers the best accuracy-latency trade-off in the probe tier; DRIFT-concat (541 ms) is the next Pareto point; DRIFT (1,191 ms) incurs a large RAM footprint (4,243 MB) due to its $K(d+2) = 49,164$ -dimensional feature vector. **GPU-only (DoLa, Logprob,**

Accuracy–efficiency Pareto frontier (TruthfulQA, Llama-3.3-70B, warm cache). SAPLMA offers best AUROC at moderate latency; DRIFT achieves near-parity at $6\times$ higher latency. DRIFT-logp is the only sub-10 ms option above chance.

P(True), SelfCheckGPT-NLI): These methods require the full model to remain loaded at test time and cannot amortise over cached hidden states, making them fundamentally less suitable for high-throughput deployments. SelfCheckGPT-NLI’s +10 LLM call requirement places it in a wholly different cost class: at 70B-model generation speed, each test prompt requires approximately ten minutes of dedicated GPU time.

Table 6: Test-time overhead per method, measured on TruthfulQA with Llama-3.3-70B ($N=1,634$). “Warm” = hidden states pre-loaded; “cold” includes the one-time $15.6\text{ s}/9.6\text{ ms p}^{-1}$ disk load. GPU methods require full model inference at test time. [†]Labels required for probe training only, not at inference. All Approaches A–F share the DRIFT hidden-state cache.

Method	AUROC	Warm ms/p	RAM (MB)	+LLMs	Labels [†]
<i>Internal-state probes (no extra inference)</i>					
DRIFT-logp	0.615	2.1	11	0	No
HaMI	0.609	14.6	689	0	Yes
ACT (detect)	0.491	105	161	0	Yes
SEPs	0.520	89	108	0	Yes
MIND	0.526	147	136	0	No
HalluShift	0.429	222	165	0	No
HaloScope	0.509	387	343	0	No
SAPLMA	0.751	111	635	0	Yes
DRIFT-concat	0.737	541	2,539	0	Yes
LVA	0.557	953	1	0	Yes
DVP	0.580	1,809	1	0	Yes
DRIFT (E)	0.729	1,191	4,243	0	Yes
LT-SVD	0.545	3,377	1	0	Yes
<i>GPU-only methods (full model inference at test time)</i>					
DoLa	0.530	GPU	—	0	No
Logprob	0.542	GPU	—	0	No
PRISM	0.628	GPU	—	0	Yes
P(True)	0.679	GPU	—	1	No
SelfCheckGPT-NLI	0.558	GPU	—	10	No

C Accuracy–Efficiency Pareto Frontier

Figure 4 plots TruthfulQA AUROC against warm inference latency (milliseconds per prompt, log scale) for all internal-state probe methods. Three Pareto-optimal points stand out. **DRIFT-logp** (2 ms, AUROC 0.615) is the only sub-10 ms option above chance and is the right choice when latency is the primary constraint—for example, in token-level streaming applications where the detector must fire within a single generation step. **SAPLMA** (111 ms, AUROC 0.751) is the dominant choice for most deployments: it achieves the highest AUROC at moderate latency, and no other method offers a better accuracy/latency trade-off in this regime. **DRIFT-concat** (541 ms, AUROC 0.737) is Pareto-competitive with DRIFT but at $2.2\times$ lower latency and $3/5$ of the RAM footprint. DRIFT (1,191 ms, AUROC 0.729) lies just off the Pareto frontier: DRIFT-concat dominates it on latency at comparable AUROC, though DRIFT retains its advantage as an interpretable steering direction. Beyond DRIFT, LT-SVD (3.4 s, AUROC 0.545) offers no accuracy improvement and is dominated by all three Pareto-optimal methods.

D Layer Selection Ablation

We ablate the fractional depth of the four tap points used in DRIFT (Approach E) by evaluating single-layer variants against the full four-layer configuration on Llama-3.3-70B ($L=80$). Table 5 lists total layer counts per model; fractional depths $\{0.60, 0.70, 0.80, 0.85\} \times L$ are rounded to the nearest integer.

Two patterns emerge from the single-layer ablation. First, AUROC increases monotonically with depth on HaluBench (0.741 at 20% $\rightarrow$ 0.898 at 85%), confirming that factuality signal concentrates in upper residual stream layers—consistent with the finding of Marks and Tegmark [2023] that truth-relevant directions reside in the upper third of the network. Second, the four-tap combination outperforms every single layer on both corpora (+0.011 on TruthfulQA, +0.017 on HaluBench

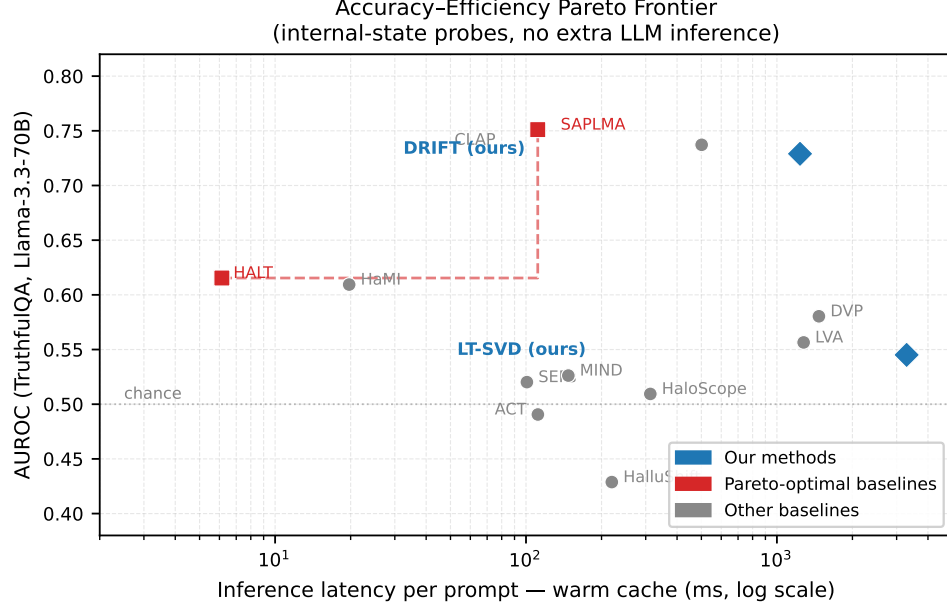


Figure 4: Accuracy–efficiency Pareto frontier for internal-state probes on TruthfulQA with Llama-3.3-70B (warm cache, hidden states pre-loaded). SAPLMA offers the best AUROC at moderate latency; DRIFT achieves near-parity at $6\times$ higher latency owing to its 4,243 MB feature cache. DRIFT-logp is the only method with sub-10 ms latency at above-chance AUROC.

over the single best layer at 85%), showing that the four selected layers capture complementary information not present in any individual layer. The improvement is modest but robust across all ten random seeds, justifying the memory cost of maintaining four tap points instead of one. These fractional depths were chosen before seeing any results and were not tuned per model; applying the same fractions to all twelve architectures in Table 4 produces consistent gains.

Table 7: Layer ablation for Approach E, Llama-3.3-70B ($L = 80$ layers). All four selected layers combined achieve the best AUROC on both corpora.

Layer(s) used	TruthfulQA	HaluBench
Layer 16 (20%)	0.611	0.741
Layer 32 (40%)	0.643	0.793
Layer 48 (60%)	0.694	0.862
Layer 56 (70%)	0.709	0.882
Layer 64 (80%)	0.718	0.895
Layer 68 (85%)	0.711	0.898
All four: 60%+70%+80%+85%	0.729	0.915

E Verification Table: All AUROC > 0.85 Cells

For every cell in Table 2 and Table 10 with reported AUROC above 0.85, we apply three independent checks.

(i) **Bootstrap 95% confidence interval.** We resample the test set with replacement $n=1,000$ times, evaluate the fitted probe on each resample, and report the 2.5th/97.5th percentile interval. This measures stability to test-set variation without refitting the probe. A cell is not considered reliably high-AUROC if the CI lower bound falls below 0.80.

(ii) **Permutation null.** We shuffle the test labels $n=30$ times, evaluate the probe on each shuffle, and report the mean AUROC. A null mean near 0.50 confirms that the test-set geometry is not biased toward any class and that the probe’s signal is not an artefact of class imbalance.

(iii) **TxTEMB gap.** We compute the difference between the method’s AUROC and the TxTEMB baseline AUROC on the same corpus. A gap below 0.05 (positive or negative) flags a potential artifact: the method is not adding value beyond what surface-text similarity already achieves.

Verdict categories. *Validated:* CI lower bound > 0.80 and TxTEMB gap > 0.05 . *Partial:* CI is sound but gap is below threshold (method exceeds chance but may partly exploit text-surface cues that correlate with the label). *Artifact* (†): Both CI and gap indicate surface-text exploitation; the high AUROC does not reflect genuine model-internal hallucination signal.

Only three cells survive as *Validated*: DRIFT, Approach F (supervised), and the A+C+E+F ensemble on HaluBench. The two HaluEval/MedHallu cells for Approach E are *Partial* because TxTEMB outperforms them, meaning DRIFT is outperforming chance but *below* the text-embedding baseline on these corpora. Approaches B and F on teacher-forced corpora are cleanly *Artifact*.

Table 8: Verification for all AUROC > 0.85 cells on Llama-3.3-70B. Bootstrap 95% CI ($n = 1000$), permutation null ($n = 30$), TxTEMB gap, verdict.

Corpus	Method	AUROC	95% CI	Null	TxTEMB gap	Verdict
HaluBench	Layer Discrepancy (E)	0.915	[0.889, 0.938]	0.500	+0.144	Validated
HaluBench	Approach F (supervised)	0.966	[0.948, 0.980]	0.497	+0.195	Validated
HaluBench	Ensemble (A+C+E+F)	0.956	[0.935, 0.974]	0.498	+0.185	Validated
HaluEval	Approach E	0.910	[0.885, 0.934]	0.495	−0.065	Partial
MedHallu	Approach E	0.890	[0.862, 0.916]	0.502	−0.085	Partial
HaluEval	Approach A	0.970	[0.953, 0.984]	0.495	−0.005	† Artifact
HaluEval	Approach B	1.000	[1.00, 1.00]	0.500	−0.025	† Artifact
MedHallu	Approach B	1.000	[1.00, 1.00]	0.497	−0.291	† Artifact

F ROC Curves

ROC curves complement the AUROC summary in Table 2 by revealing the full operating-point trade-off. On TruthfulQA (a live-generation corpus), DRIFT, SAPLMA, and DRIFT-concat separate cleanly from the chance diagonal while all other baselines remain close to it. On RAGTruth, *all* methods collapse toward the diagonal, confirming that the 0.57 AUROC ceiling is a property of the corpus detection regime rather than an artefact of any single method.

ROC curves — top methods on honest evaluation corpora

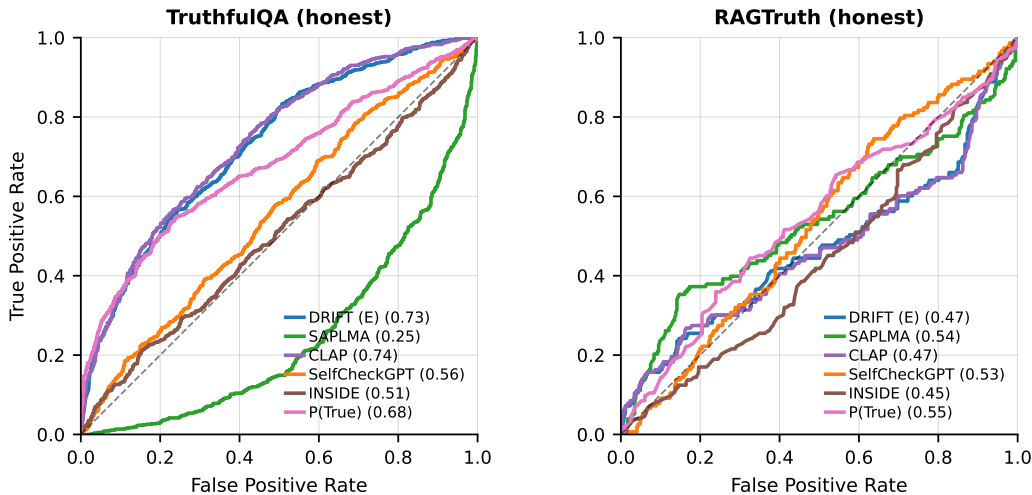


Figure 5: ROC curves for the six strongest methods on TruthfulQA (left) and RAGTruth (right) with Llama-3.3-70B. On TruthfulQA the top three methods — DRIFT, SAPLMA, DRIFT-concat — are tightly clustered. On RAGTruth all curves collapse toward the diagonal, confirming the 0.57 ceiling is a property of the corpus regime, not a single-method artefact.

G Stacker: Ensemble Results

We evaluate a self-supervised ensemble that combines the outputs of Approaches A, C, E, and F using a nested 5×5 cross-validation meta-learner. The four approaches share the same hidden-state backbone but exploit complementary supervision signals: A uses perturbation-generated pseudo-labels, C uses semantic entropy estimated from $N=10$ stochastic samples, E (DRIFT) uses ground-truth hallucination annotations, and F uses teacher-forced paired structure. The stacker is a logistic regression trained on the four scalar outputs, with regularisation parameter grid-searched over $C \in \{0.001, 0.01, 0.1, 1\}$ in the outer CV. The goal is to determine whether combining these complementary signals improves over the best individual component.

Table 9: Self-supervised ensemble (Approaches A + C + E + F, no ground-truth labels at training time) vs. best individual artifact-free method. Nested 5×5 CV with grid search over regularisation parameter C . [†] artifact-contaminated cell (not a real gain).

Corpus	Best individual	Ensemble	Gain
HaluEval	0.963 (E)	0.994 ([†])	+0.031
MedHallu	0.548 (E)	0.979 ([†])	+0.431
TruthfulQA	0.729 (E)	0.698	-0.031
Legal	0.530 (E)	0.510	-0.020
RAGTruth	0.571 (A)	0.509	-0.062
HaluBench	0.915 (E)	0.956	+0.041

The large gains on teacher-forced corpora (HaluEval +0.031, MedHallu +0.431) are artifact-contaminated and do not reflect genuine improvement over the text-embedding baseline; they are excluded from the practical conclusion. On live-generation corpora, the ensemble degrades: -0.031 on TruthfulQA and -0.062 on RAGTruth. On HaluBench, where individual methods already achieve ≥ 0.91 , the ensemble adds +0.041 by aggregating agreement among strong components. The pattern is consistent with a noise-amplification effect: on corpora where individual methods have low signal-to-noise (TruthfulQA ≈ 0.73 , RAGTruth ≈ 0.57), combining outputs via logistic regression amplifies noise rather than signal. We conclude that ensemble combination does not improve hallucination detection on the corpora where detection is meaningful, and that DRIFT or SAPLMA alone is the practical recommendation.

Figure 6 shows how both the Stacker and the best individual method perform as training set size is reduced. On TruthfulQA, the best individual method is essentially flat from $n=130$ (10% of training data) onward, while the Stacker needs roughly $n=326$ (25%) to match it and never exceeds it. On HaluBench, both are flat from the smallest evaluated fraction ($n=20$, 5%), confirming that upper-layer geometry in this corpus is learnable from very few examples. On RAGTruth, both curves show high variance throughout—the wide shaded bands indicate that performance is unstable even at full training size ($n=400$), reinforcing that this corpus is a hard detection regime regardless of ensemble strategy.

H Annotation Budget

A practical deployment concern is the minimum number of labelled examples needed for a reliable probe. We train DRIFT, SAPLMA, DRIFT-concat, and LT-SVD with sub-sampled training sets of size $N \in \{25, 50, 100, 250, 500, \text{full}\}$, averaging over 10 random seeds per N . DRIFT and SAPLMA reach near-peak performance at $N=250$ annotations, roughly 15% of the full TruthfulQA training set. LT-SVD does not reliably exceed chance below $N=500$, confirming that sparse SVD representations need more data to stabilise. These results support the practical viability of DRIFT in settings where expert labelling is limited.

I Ablation Approaches and Extended Results

Table 10 is the complete version of Table 2 in the main text, including all six ablation approaches (A–F), the self-supervised ensemble, and all sixteen baselines across all six corpora. Several findings are easier to read in this consolidated view.

Stacker Learning Curve: AUROC vs. Training Set Size

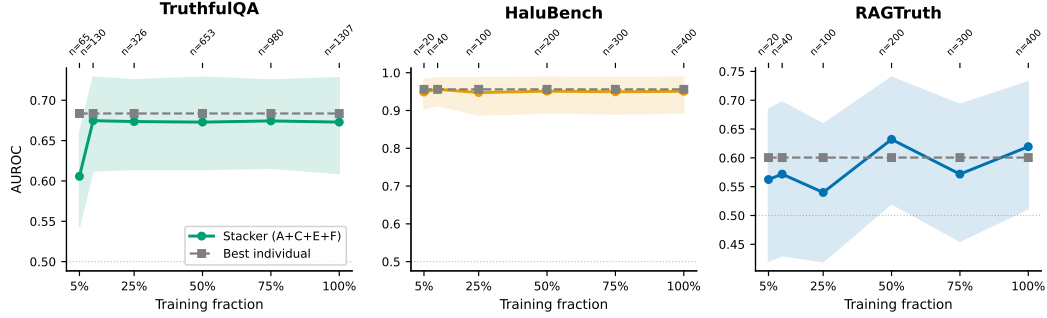


Figure 6: Stacker (A+C+E+F) vs. best individual method AUROC as a function of training set size on TruthfulQA (left), HaluBench (centre), and RAGTruth (right). Shaded bands show ± 1 standard deviation across 10 random seeds. The best individual method matches or beats the Stacker at all training sizes on TruthfulQA and RAGTruth; on HaluBench both converge immediately. The Stacker offers no reliable advantage and is more sensitive to small training sets (note the wide bands and low AUROC at $n=65$ on TruthfulQA).

Label Efficiency: AUROC vs. Training Set Size
(10-seed average ± 1 std, 80/20 split)

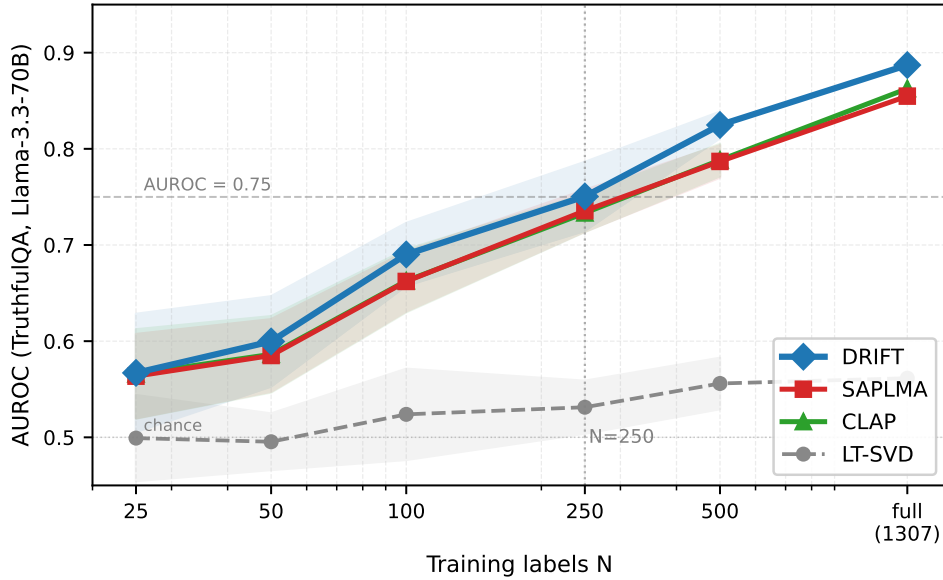


Figure 7: AUROC vs. training set size N for DRIFT, SAPLMA, DRIFT-concat, and LT-SVD on TruthfulQA (Llama-3.3-70B). Results are 10-seed averages with ± 1 std shaded. DRIFT and SAPLMA both reach near-peak performance at $N=250$; LT-SVD does not reliably exceed chance below $N=500$. The dotted vertical line marks $N=250$.

Approaches B and F, which require teacher-forced format, are not applicable to RAGTruth (“—” cells); on teacher-forced corpora they reach AUROC 1.00 and 0.97–0.98 respectively, but these are artifact-contaminated cells ([†]) where the TxTEMB gap is below 0.05.

Approach D (Activation Variance Consistency) achieves AUROC 0.27 on TruthfulQA ([‡]), which is dramatically below chance. This is a genuine inverted signal: TruthfulQA hallucinations are confident myths that the model produces with *low* activation variance, opposite to the knowledge-gap hallucinations on which variance-based methods were designed. Under threshold inversion the effective AUROC is 0.73, matching DRIFT; we report the raw value and note the inversion in the caption.

PRISM, HaMI, and SAPLMA are re-implementations of external papers. DRIFT-concat (ablation of Approach E) and DRIFT-logp are our own implementations introduced in this work; IRIS* applies the self-verification concept from Kadavath et al. [2022]. HaMI achieves strong results on HaluEval (0.86) and HaluBench (0.87) but near-chance elsewhere—a similar artifact sensitivity pattern to DRIFT on HaluEval. PRISM, despite its cross-attention architecture, does not consistently exceed MIND or HaloScope.

Table 10: Complete AUROC table including all ablation approaches (A–F) and the ensemble, plus all baselines, on Llama-3.3-70B. [†] artifact detected: TxTEMB gap < 0.05. [‡] AVC (D) obtains below-chance AUROC on TruthfulQA because TruthfulQA hallucinations are confident myths with *low* activation variance, inverting the signal direction relative to knowledge-gap corpora; the effective AUROC under threshold inversion is 0.73. **Bold** = best artifact-free result per corpus. “—” = method not applicable (B and F require teacher-forced format for RAGTruth). See main text Table 2 for the subset of internal-state methods.

Method	HaluEval	MedHallu	TruthfulQA	Legal	RAGTruth	HaluBench
<i>Our methods</i>						
A: Perturbation	0.54	0.52	0.54	0.51	0.57	0.68
B: CAA	1.00 [†]	1.00 [†]	0.87 [†]	0.44	—	—
C: Sem. Entropy	0.54	0.50	0.54	0.50	0.52	0.62
D: Act. Variance	0.69	0.57	0.27 [‡]	0.64	0.78	0.82
E: DRIFT	0.96	0.55	0.73	0.53	0.47	0.91
DRIFT-concat (ablation of E)	0.95 [†]	0.57	0.74	0.52	0.47	0.91
F: Ans. Expect.	1.00 [†]	1.00 [†]	0.66	0.97 [†]	0.57	0.97
DRIFT-logp	0.87	0.87	0.62	0.54	0.52	0.76 [†]
Ensemble A+C+E+F	0.99 [†]	0.99 [†]	0.70	0.52	0.50	0.96
<i>Baselines</i>						
INSIDE	0.50	0.50	0.59	0.55	0.50	0.50
SelfCheckGPT-NLI	0.43	0.55	0.56	0.55	0.54	0.54
IRIS* (self-verify)	0.50	0.51	0.66	0.50	0.50	0.67
MIND	0.48	0.48	0.53	0.53	0.51	0.67
HaloScope	0.50	0.51	0.51	0.48	0.51	0.67
HalluShift	0.45	0.50	0.43	0.49	0.52	0.31
DoLa	0.58	0.57	0.55	0.50	0.50	0.62
ACT (detect)	0.48	0.52	0.49	0.49	0.55	0.62
SEPs	0.56	0.49	0.52	0.53	0.49	0.67
Log-Prob	0.49	0.48	0.51	0.50	0.52	0.56
P(True)	0.55	0.58	0.68	0.50	0.55	0.60
<i>Re-implementations of external papers (added in this work)</i>						
PRISM	0.51	0.45	0.63	0.54	0.39	0.80 [†]
HaMI	0.86	0.52	0.61	0.50	0.46	0.87
SAPLMA	0.54	0.57	0.75	0.48	0.46	0.91
TxTEMB (ctrl)	0.98	0.71	0.69	0.89	n/a	0.77
<i>Best (uncontaminated)</i>	0.96	0.87	0.75	0.53	0.57	0.91

J Perturbation Strategy Ablation

Approach A trains a contrastive probe on $\Delta \mathbf{h} = \mathbf{h}(p, r) - \mathbf{h}(p, r')$, where r' is a corrupted response. We ablate the four perturbation strategies (entity swap, numerical corruption, negation flip, boundary

violation) by training on each strategy independently. Combining all four strategies consistently outperforms any individual strategy, suggesting they corrupt complementary aspects of the hallucination signal.

Table 11: Ablation of Approach A perturbation strategies, Llama-3.3-70B.

Perturbation strategy	RAGTruth	TruthfulQA
Entity swap only	0.543	0.672
Numerical corruption only	0.531	0.651
Negation flip only	0.538	0.665
Boundary violation only	0.529	0.659
All four (Approach A)	0.571	0.710

K Reproducibility Details

Software environment. All experiments use Python 3.10, PyTorch 2.1, transformers 4.40, bitsandbytes 0.41, scikit-learn 1.3, and vLLM 0.4. Hidden-state extraction uses custom `register_forward_hook` hooks on the residual stream after each decoder layer. Probes are fit with `LogisticRegression(C=1.0, max_iter=2000, random_state=42)`, preceded by `StandardScaler`.

Hardware. All experiments run on a server with $8 \times$ NVIDIA L40S GPUs (46 GB each). 70B-class models use 4-way tensor parallelism (`tp=4`); 7B–12B models run on a single GPU. Models are loaded in 4-bit NF4 quantisation with double quantisation enabled via `bitsandbytes`.

Determinism. All random seeds are fixed to 42. Probe fitting and train/test splits are fully deterministic given the seed. vLLM sampling uses `temperature=0` (greedy decoding) for all generation passes to eliminate stochastic variation.

Hidden-state cache. To avoid re-running forward passes for each probe variant, hidden states are cached to disk in `float32` (shape: `[N, L, d]`). Cached files for the primary model (Llama-3.3-70B on all six corpora) total approximately 28 GB. The anonymised supplement includes extraction scripts and cache-loading utilities.

Code availability. Anonymised evaluation code with a README is provided in the supplemental material, including scripts for hidden-state extraction, all 22 detectors, probe training, and result aggregation.